Joe Worsfold

Founder & Principal Architect

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Summary

Founder and Technical Director at Stack Foundry UK. Principal Architect for agentic AI workflows and event-driven systems in Rust, TypeScript, and Python. AWS AI Award winner. I embed as a principal technical leader — or consult as an external architect — to ship autonomous, high-performance systems without sacrificing operational reliability.

Skills

Core: Rust, TypeScript, Python, Go, Agentic AI workflows, RAG & embeddings, AWS (serverless / event-driven)

Also strong: OpenSearch, LangChain agents, Neo4j, DynamoDB / Postgres, React / Next.js, CDK / Terraform, Cursor / AI-native engineering

Experience

Founder & Principal Architect, Stack Foundry Ltd 07/2025 – Present1 year 4 months



Remote

Founder-led consultancy. Principal architecture for agentic AI workflows and specialty-insurance data systems; building Prism and Helmsman.

Consultancy started while wrapping Aviva Zero.

RustTypeScriptPythonAWS (LambdaOpenSearchevent-driven)LangChainRAG/embeddingsSurrealDBCursor

- **Production MDM platform:** Delivered a production master data management platform for a specialty insurance client in six months, replacing multi-system complexity with one source of truth — fewer engineers to maintain and fewer ops agents to run. Used at production scale by underwriters and dataops.
- **RAG & semantic search:** Designed LLM selection, RAG, and embedding pipelines with OpenSearch semantic enrichment to improve discoverability across messy insurance master data.
- **Agentic entity resolution:** When deterministic entity resolution failed, introduced LangChain agentic workflows so the system could gather context and propose decisions for human approval — increasing agency without silent autonomous binds.
- **AI-native engineering:** Taught client developers AI-native delivery and built bespoke guardrails for agentic coding workflows (Cursor-centred practice).

Senior Full Stack Software Engineer, Aviva Zero 01/2024 – 11/20251 year 10 months



London

Climate-focused insurance platform — high-availability Go/TypeScript services and accessible React frontends.

Backend: Go, AWS Lambda, API Gateway, EventBridge, SQSFrontend: TypeScript, React, Next.js, ECSCI/CD: CDK, GitHub Actions, Datadog, PagerDuty

- **Event-driven serverless:** Designed and implemented event-driven serverless architectures in Go using Lambda, SQS, EventBridge, and SES for asynchronous workflows, scheduling, and alerting across policy and payment services.
- **Scale & availability:** Built and maintained a platform that has sold over a million policies and supports tens of thousands of daily users, with 24/7 reliability through monitoring and performance work.
- **Financial services:** Delivered financially sensitive services including BACS payment workflows, interest-bearing monthly instalments, and secure customer interactions within FCA-regulated boundaries.
- **Observability:** Led alerting and observability improvements for proactive detection and resolution.

Lead Software Engineer (Tech Lead), Flock Cover 02/2023 – 01/202411 months



London

Next-generation insurance platform with modern cloud and serverless infrastructure.

Backend: TypeScript, Scala, Lambda, API Gateway, EventBridge, SQSFrontend: React, Next.js, ECS/LambdaIaC: CDK, Terraform, GitHub Actions, Datadog

- **Insurance platform:** Led full-stack development across Node/TypeScript backends and React/Next.js frontends with a focus on scalability and performance.
- **Microservices:** Designed microservices on Lambda, ECS, and EventBridge for observable, 24×7 services that scale with demand.
- **Infrastructure as Code:** Championed Terraform and CDK for repeatable AWS provisioning.

Senior Consultant Full Stack Engineer, Infinity Works (part of Accenture) 10/2021 – 02/20231 year 4 months



UK

Tech lead for specialty-insurance automation and serverless integrations (client work anonymised where required). TypeScriptNode.jsPostgresDynamoDBLambdaAPI GatewayStep FunctionsEventBridgeTerraformDatadog

- **Underwriting automation:** Led automation for underwriting data extraction and distribution with TypeScript, Node.js, and Postgres on AWS.
- **Event-driven workflows:** Built serverless, event-driven integrations with Step Functions and EventBridge for multi-service workflows.
- **Tech leadership:** Acted as Tech Lead — unblocked hard engineering problems, ran knowledge-sharing, and strengthened engineering culture.

Software Engineer, UK Home Office 09/2020 – 10/20211 year 1 month



UK

Graph-backed intelligence tooling and government web applications.
ReactAngularNode.jsMongoDBNeo4jDockerKubernetesDrone

- **Graph intelligence:** Supported ETL into Neo4j graph databases modelling people, assets, and movements; wrote performant Cypher for pattern discovery at scale — central to the platform's intelligence capabilities.
- **Operationalising ML:** Worked with data scientists to operationalise ML against the graph layer for anomaly detection and higher-quality investigative leads.
- **Government applications:** Built React/Angular apps and Node REST APIs with TDD/BDD, Docker, and Kubernetes CI via Drone.

IT Developer, Mott MacDonald 09/2018 – 09/20202 years



UK

ERP integration and process automation.
JavaJavaScriptPL/SQL

- **ERP integration:** Built Java integrations from legacy systems to cloud ERPs using SOLID, design patterns, and TDD.
- **Process automation:** Delivered electronic forms/workflows and PL/SQL-backed data solutions.

Selected projects

Tooling Helmsman

Open rules, skills, and checks for agent-written code — readiness scoring, review, and STAR-stack guardrails. Used on Prism and NIDOS.

Context

Hack Hate 2020: Hate Detector

Winner — Best use of AI/ML (AWS) and Best use of location (ESRI). Processed and classified 18.5M+ tweets; graph model in Neo4j for hate-speech network visualisation.

Presentation · Code

Education

Birkbeck, University of London 2017 – 2019
MSc Computer Science, Distinction

Algorithms, data structures, machine learning, and NLP with emphasis on scalable AI-driven systems.

Thesis: Character-level Convolutional Neural Networks for Hate Speech Detection with Intelligent Adversaries — CNN text classification at scale with graph modelling of classified content in Neo4j.

Royal Holloway, University of London 2009 – 2013
BA History, 2:1

Analytical writing, research, and communication — foundations that still show up in product and stakeholder work.